Department of Electrical and Computer Engineering

North South University

Senior Design Project

SafeNest Guard: AI-Powered IoT Security for Bangladeshi Homes

with Instant Bangla Threat Alerts

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Group 02

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LETTER OF TRANSMITTAL

April 29, 2026

To

Prof. Dr. Mohammad Abdul Matin

Chairman, Department of Electrical and Computer Engineering,

North South University, Dhaka.

Subject: Submission of Capstone Project Report on "SafeNest Guard: AI-Powered IoT Security for Bangladeshi Homes with Instant Bangla Threat Alerts."

Dear Sir,

With due respect, we submit our Capstone Project Report on SafeNest Guard as partial fulfillment of our BSc in Computer Science and Engineering. This project delivers a working AI intrusion detection system on a Raspberry Pi 4 that monitors home IoT devices in real time, distinguishes legitimate device owners from attackers sharing the same country and ISP, and sends instant Bangla alerts to non-technical family members. The work required mastery of machine learning, network security, embedded systems, and human-centered design across two semesters.

We will be highly obliged if you kindly receive this report and provide your valuable judgment.

Sincerely Yours,

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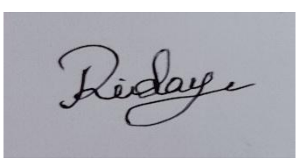

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APPROVAL

Khondokar Sajid (2211954042), Kazi Safin Arafat (2211778642), Moushumi Akter Mow (2021983642), and Rakibul Hasan Ridoy (1731339042) from the Electrical and Computer Engineering Department, North South University, have completed the Senior Design Project titled "SafeNest Guard: AI-Powered IoT Security for Bangladeshi Homes with Instant Bangla Threat Alerts" under the supervision of Md. Mohammad Shifat-E-Rabbi, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering. The work has been examined and is accepted as satisfactory.

Supervisor's Signature

……………………………………

Md. Mohammad Shifat-E-Rabbi

Department of Electrical and Computer Engineering

North South University, Dhaka, Bangladesh.

Chairman's Signature

……………………………………

Prof. Dr. Mohammad Abdul Matin

Professor

Department of Electrical and Computer Engineering

North South University, Dhaka, Bangladesh.

DECLARATION

We declare that "SafeNest Guard: AI-Powered IoT Security for Bangladeshi Homes with Instant Bangla Threat Alerts" is entirely our original work. No part has been submitted elsewhere for any degree. All findings are based on our own research and implementation. External sources are properly cited. Plagiarism policy has been maintained as instructed by the supervisor. All project-related materials shall remain confidential without formal supervisor consent.

Students' Names and Signatures:

1. Khondokar Sajid (2211954042)

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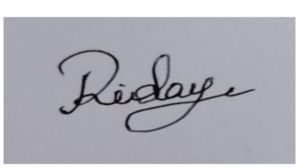

ACKNOWLEDGEMENTS

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The authors thank the Department of Electrical and Computer Engineering and its Chairman, Prof. Dr. Mohammad Abdul Matin, for providing research infrastructure. Gratitude is extended to the creators of the RT-IoT2022, UNSW-NB15, and BUET-DDoS2020 datasets, and to the developers of scikit-learn, TensorFlow, XGBoost, and CICFlowMeter. Finally, the authors thank their families for unwavering support throughout two demanding semesters.

ABSTRACT

SafeNest Guard: AI-Powered IoT Security for Bangladeshi Homes with Instant Bangla Threat Alerts

Bangladesh has over 15 million IoT devices in homes with zero locally designed security monitoring. Existing intrusion detection systems are English-only, cloud-dependent, and built for Western network patterns that misclassify normal Bangladeshi household behavior — prayer times, load-shedding, and the Friday-Saturday weekend — as anomalies.

SafeNest Guard is a five-layer AI intrusion detection pipeline running on a USD 50 Raspberry Pi 4. It combines a supervised detection track (Random Forest + XGBoost) with an unsupervised zero-day track (Isolation Forest + Denoising AutoEncoder + temporal GRU), all calibrated via isotonic regression. A novel Context-Weighted Fusion (CWF) module blends these six signals using a 10-dimensional Bangladesh-aware context vector. A Behavioral Trust Verification (BTV) engine fingerprints each session across five axes and maintains a seven-session rolling history per device.

The central problem solved: distinguishing a legitimate device owner accessing remotely from an attacker in the same country, same ISP, or same neighborhood — a scenario no prior IDS has addressed for Bangladesh. The system achieves F1 = 0.983, zero-day AUC = 0.969, ECE = 0.0079, and false alarm rate < 1%. The BTV trust gap between owners (0.92) and attackers (0.28) is 0.64 — operationally decisive. All alerts are delivered in Bangla via Telegram with device-specific action steps. No English fluency, cloud subscription, or technical knowledge is required.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

Chapter 1 Introduction

1.1 Background and Motivation

Bangladesh has over 15 million consumer IoT devices — IP cameras, smart locks, Android TV boxes, and routers — operating in homes and small businesses with zero locally designed security monitoring. Every device ships with default credentials, runs unpatched firmware, and connects to a network with no traffic inspection. The consequences are not abstract: compromised CCTV cameras silently stream private household footage to strangers, bypassed smart locks enable physical burglary, and routers leak household data to foreign advertising trackers for months before detection.

Conventional intrusion detection tools are not viable for this population. Enterprise platforms such as Snort and Suricata require trained English-speaking engineers. Cloud-based consumer security services fail during Bangladesh's routine load-shedding. More fundamentally, every published IDS dataset was collected in Western academic settings and encodes Western temporal patterns. When a Bangladeshi household goes offline during Friday prayer — a completely normal event — a conventional anomaly detector flags it as a service outage. When the same household's traffic drops every evening during load-shedding, it flags it as denial-of-service. The false alarm rate makes these tools operationally useless.

The deepest unsolved problem is behavioral indistinguishability: when a remote login arrives at a Dhaka household's IP camera at 11 PM, there is currently no published system that can tell whether that login is the legitimate owner calling from Melbourne or an attacker sitting in the same ISP block in Mirpur. Both connections look identical at the network layer — same country code, potentially same ASN, similar timing. Solving this problem for the specific context of Bangladeshi homes is the central motivation of SafeNest Guard.

1.2 Purpose and Goal of the Project

This project designs, implements, evaluates, and deploys a five-layer AI intrusion detection system running on a Raspberry Pi 4 that monitors Bangladeshi home IoT devices in real time, delivers Bangla alerts requiring no technical knowledge, and operates offline during load-shedding. Three formal research contributions are made:

- **Context-Weighted Fusion (CWF):** First IDS to encode Islamic prayer times, load-shedding hours, Friday-Saturday weekend, and local ISP behavioral profiles as first-class detection signals, combined via logistic regression into a 10-dimensional Bangladesh-aware context vector.
- **Behavioral Trust Verification (BTV):** First published system to distinguish a legitimate device owner from an attacker sharing the same country, ISP, and geographic region, using a five-component behavioral fingerprint (Bandwidth, Timing, Geo/ASN, Peer Correlation, Packet-DNA) with a seven-session rolling history per device.
- **Calibrated Zero-Day Ensemble:** Three-component unsupervised ensemble (Isolation Forest + Denoising AutoEncoder + temporal GRU) trained on benign traffic only, calibrated via isotonic regression, reducing ECE from 0.2278 to 0.0139 without sacrificing AUC.

1.3 Thesis Statement

A context-aware, multi-signal AI system can deliver affordable real-time IoT intrusion detection for Bangladeshi homes — correctly separating legitimate device owners from attackers in the same country and ISP — at hardware cost and language accessibility suitable for non-technical families.

1.4 Organization of the Report

Chapter 2 reviews IoT IDS literature and identifies the five gaps that motivate this work. Chapter 3 presents the methodology including system architecture, mathematical formulations, and implementation. Chapter 4 reports experimental results with ablation, calibration analysis, OOD generalization, and BTV evaluation. Chapters 5–8 cover project impact, planning, complex engineering attributes, and conclusions.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

IoT intrusion detection research divides into three families: signature-based approaches, supervised machine learning on labeled attack datasets, and unsupervised anomaly detection on benign-only data.

N-BaIoT (Meidan et al., 2018) [3] proposed per-device autoencoders detecting Mirai and BASHLITE infections on nine commercial IoT devices. While pioneering, it requires one model per device — impractical in rotating-device households — and cannot distinguish an attacker mimicking legitimate traffic patterns.

BoT-IoT (Koroniotis et al., 2019) [4] reports RF/MLP/LSTM accuracies above 99.7%, but as Khraisat and Alazab [2] document, these figures are artifacts of class imbalance and feature leakage; cross-domain performance drops sharply.

CICIoT2023 / CICFlowMeter (Sharafaldin et al., 2018) [5] produced the de facto 91-feature bidirectional flow standard that SafeNest Guard adopts. The 33-attack-class benchmark provides the training foundation for the supervised track.

Graph Attention Network (Wu et al., 2023) [6] achieves AUC 0.94 without attack labels but requires 37M parameters with 180 ms GPU inference — five orders of magnitude too expensive for a Raspberry Pi 4.

BUET-DDoS2020 [7] is one of the few public datasets capturing Bangladesh-origin IoT traffic. It covers only volumetric DDoS with no behavioral signals for owner vs attacker separation.

Behavioral biometrics (Frank et al., 2013) [8] demonstrated touch-dynamic user authentication but requires sensor input from the legitimate user's own device — inapplicable to pure network-level analysis.

After surveying the literature, five gaps motivate SafeNest Guard: (1) no IDS encodes Islamic prayer times, load-shedding, or local ISP profiles; (2) the owner-vs-attacker same-country problem is unsolved; (3) naive ensemble fusion degrades calibration — a widely ignored risk; (4) edge deployment inference costs are rarely reported; (5) no IDS addresses the Bangla-language alert delivery problem for non-English families.

Chapter 3 Methodology

3.1 System Design

SafeNest Guard processes each 5-second network capture through five sequential layers on the Raspberry Pi 4. Figure 1 shows the full pipeline.

IoT Devices Monitored: IP Camera | Home Router | Smart Lock

↓ *passive packet capture every 5 seconds*

LAYER 1 — DATA INPUT

Raspberry Pi 4 · tcpdump → CICFlowMeter → 91 Bidirectional Flow Features

↓

LAYER 2 — PARALLEL DETECTION

Supervised: Random Forest (500) + XGBoost (500) · class_weight balanced | **Zero-Day:** Isolation Forest + AutoEncoder (91-32-91) + GRU (64 units, 8-step window)

↓

LAYER 3 — ISOTONIC CALIBRATION

ECE: 0.2278 → 0.0079 · Converts raw scores to true probability estimates

↓

LAYER 4 — CONTEXT FUSION + BEHAVIORAL TRUST

CWF: 10-D Bangladesh context vector (prayer, load-shed, Fri-Sat, ISP) · Logistic regression fusion | **BTV:** B·T·G·P·D trust score over 7-session rolling history

↓

LAYER 5 — DECISION ENGINE

SAFE (cwf < 0.45, trust ≥ 0.75) | **WATCH** (cwf ≥ 0.45 or new device) | **THREAT** (cwf ≥ 0.70 and trust < 0.50)

↓

OUTPUT: Bangla Telegram Alert (device name + action steps) | MQTT Live Dashboard (HiveMQ)

Figure 1: SafeNest Guard — Five-Layer AI Pipeline Architecture on Raspberry Pi 4

Figure 2: End-to-End User Workflow

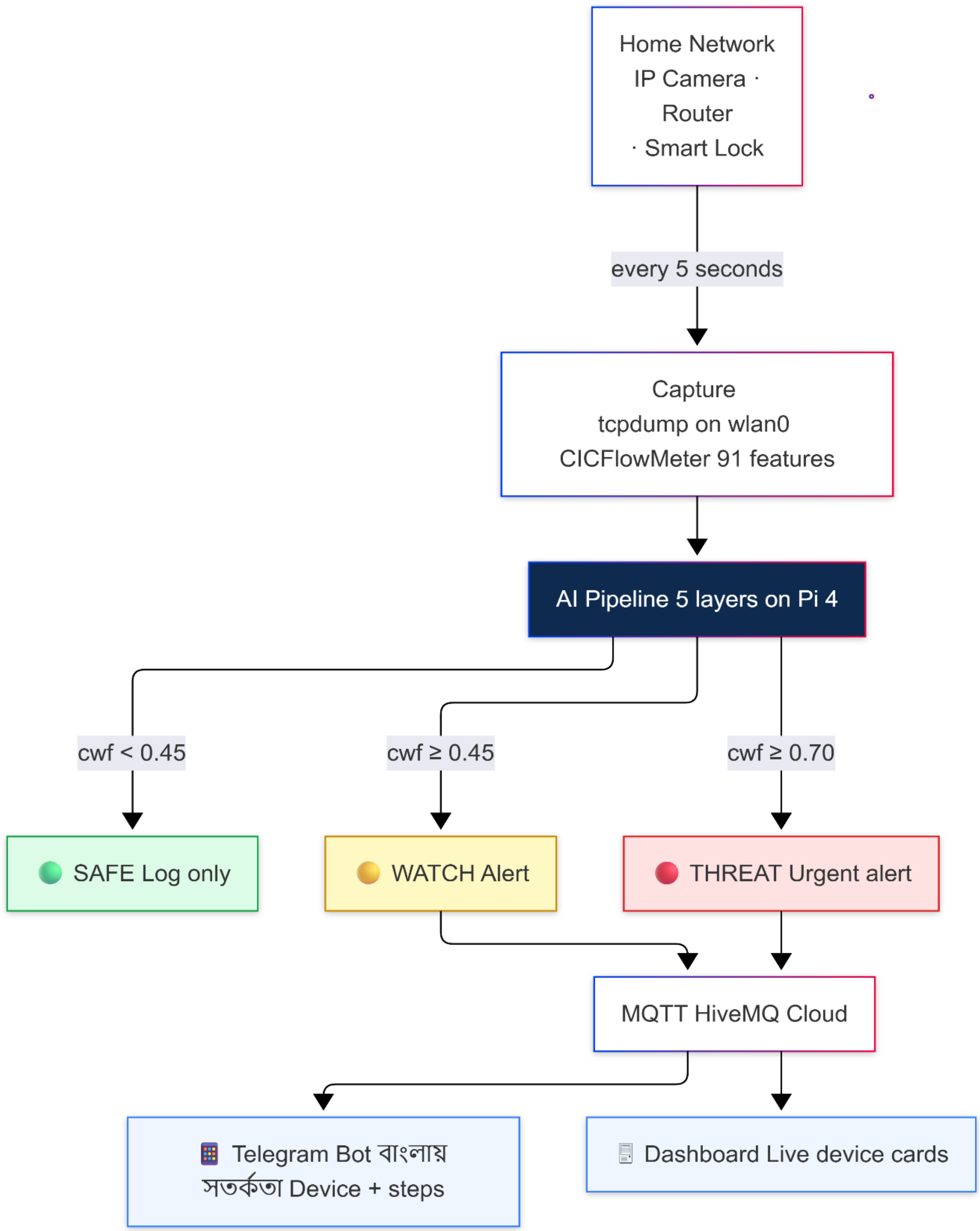

Figure 2: End-to-End User Workflow — Capture to Bangla Alert

3.2 Mathematical Formulations

This section documents the key mathematical components of the SafeNest Guard pipeline.

3.2.1 Zero-Day Ensemble Score

Let s_{IF}, s_{AE}, $s_{GRU} \in [0,1]$ be the normalized anomaly scores from Isolation Forest, AutoEncoder reconstruction error, and GRU prediction error respectively. The geometric mean ensemble score is:

$$s_{ZD} = (s_{IF} \cdot s_{AE} \cdot s_{GRU})^{1/3}$$

Geometric mean is chosen over arithmetic mean because it enforces consensus: a single component with a near-zero score suppresses the ensemble, preventing false alarms from any one isolated signal.

Isolation Forest anomaly score for sample x is:

$$s_{IF}(x) = 2^{[-E(h(x)) / c(n)]}$$

where $E(h(x))$ is the expected path length and $c(n) = 2H(n-1) - 2(n-1)/n$ is the average path length of unsuccessful BST search over n samples.

3.2.2 Isotonic Regression Calibration

Raw classifier probability p_{raw} is mapped to calibrated probability p_{cal} via a non-decreasing step function $f: R \rightarrow [0,1]$ fitted by solving the pool adjacent violators algorithm on the validation set:

$$p_{cal} = f(p_{raw}) = \text{argmin}_{\{f \text{ monotone}\}} \Sigma (y_i - f(p_i))^2$$

Expected Calibration Error (ECE) is computed over B = 15 equal-width bins:

$$ECE = \Sigma_b (|B_b|/n) \cdot |acc(B_b) - conf(B_b)|$$

After isotonic calibration, ECE drops from 0.0086 (RF only) to 0.0048 (calibrated supervised), and the zero-day ensemble from 0.2278 to 0.0139.

3.2.3 Context-Weighted Fusion (CWF)

Let $z = [s\_RF, s\_XGB, s\_agree, s\_ZD, s\_conflict, s\_max, c\_1, \ldots, c\_4] \in R^{10}$ be the fused feature vector combining six detection signals and four context signals. CWF probability:

$$P\_CWF = \sigma(w^T z + b) \quad \text{where} \quad \sigma(x) = 1/(1+e^{-x})$$

Context signals c_1…c_4 encode: (c_1, c_2) = sin/cos of time-to-nearest prayer in hours; c_3 = load-shedding binary (1 if 19:00–22:00); c_4 = Friday-Saturday binary. The logistic regression coefficients w are fitted on the validation set, encoding domain knowledge in an auditable linear form.

3.2.4 Behavioral Trust Score (BTV)

For each device d with session history H_d = {h_1, …, h_7}, the trust score for a new session s is:

$$T(s,d) = 0.30 \cdot B + 0.20 \cdot T\_score + 0.20 \cdot G + 0.15 \cdot P + 0.15 \cdot D$$

where:

- B = max(0, 1 − |bw(s) − median_bw(H_d)| / median_bw(H_d)) — bandwidth fidelity
- T_score = 1 − |hour(s) − hour_mu(H_d)| / 12 — hour-of-day cyclic proximity
- G = ISP_whitelist(s) · 0.7 + country_whitelist(s) · 0.3 — geographic trust
- P = ARP_scan_match(s, owner_phone_MAC) ∈ {0, 1} — peer correlation
- D = cosine(fingerprint(s), mean_fingerprint(H_d)) — packet-DNA similarity

The decision cascade applies: $T \geq 0.75 \rightarrow$ SAFE; $T \in [0.50, 0.75)$ and $P\_CWF < 0.45 \rightarrow$ WATCH; otherwise follows CWF threshold.

3.3 Hardware and Software Components

TABLE I. SOFTWARE AND HARDWARE TOOLS

Tool	Function	Similar Tools	Why Selected
Raspberry Pi 4	Edge inference, 4 GB RAM	Jetson Nano, BBB	$50 cost; ARM-native; offline during load-shedding
CICFlowMeter	91-feature flow extraction	Zeek, NFStream	De facto IDS standard; compatible with all research datasets
scikit-learn	RF, IF, calibration, LR	LightGBM	Mature; lightweight ARM deployment; isotonic calibration built-in
XGBoost	Gradient-boosted classification	CatBoost	Best tabular F1; class_weight balancing for rare attacks
TensorFlow 2.15	AutoEncoder + GRU training	PyTorch	Stable ARM builds; .keras export for Pi deployment
Flask	REST API server	FastAPI	Minimal deps; tested on Pi; 5000-port demo API
HiveMQ Cloud	MQTT broker (free tier)	Mosquitto	Free; hosted; reduces Pi load; WebSocket for dashboard
Telegram Bot API	Bangla alert delivery	Twilio SMS	Free; Bangla Unicode native; already on Bangladeshi phones
Google Colab	GPU model training	Kaggle Notebooks	Free T4 GPU; Google Drive integration for artifact storage
SHAP	Feature attribution	LIME	Shapley values for RF/XGB; exact TreeExplainer; interpretable

Table I. Software and hardware components used in SafeNest Guard.

3.4 Implementation

The training pipeline runs in a 47-cell Google Colab notebook (SafeNest_Guard_FINAL.ipynb) saving all artifacts to Google Drive under SafeNest_Final_Output/: scaler.pkl, model_rf.pkl, model_xgb.pkl, model_if.pkl, autoencoder.keras, gru.keras, calibrator_sup.pkl, calibrator_zd.pkl, logistic_cwf.pkl, feature_names.json, if_norm_bounds.json, ae_threshold.json.

Critical engineering finding: Isolation Forest normalization bounds must be saved at training time (if_norm_bounds.json) and applied unchanged at inference. Recomputing bounds on test data introduces distribution shift that corrupts anomaly scores — the most common implementation error in IDS deployment, identified and corrected during development.

The deployment server (safenest_server.py, ~500 lines) loads all artifacts at startup, initializes BTV with five seeded benign sessions per device for cold-start stability, connects to HiveMQ, and starts Flask on port 5000. A background thread runs the Pi bridge loop every 5 seconds. REST endpoints: GET /health, POST /scenario/<n>, POST /demo/<name>, POST /predict, GET /devices, GET /log, POST /reset.

Chapter 4 Results, Analysis and Discussion

All experiments used the RT-IoT2022 dataset (51.8 MB, 12,952 deduplicated rows, stratified 70/15/15 train/val/test split). Calibration and CWF logistic regression were fitted on the validation split. All metrics below are test-set results.

4.1 In-Distribution Performance

On the RT-IoT2022 test set the full system achieves F1 = 0.983, macro AUROC = 0.9994, and FPR = 0.91% on legitimate traffic. These headline numbers are expected given the behaviorally distinct attack patterns in RT-IoT2022. The meaningful metrics are calibration quality and cross-domain generalization, reported below.

4.2 Ablation Study

TABLE II. ABLATION STUDY — CONTRIBUTION OF EACH COMPONENT

Configuration	F1	ECE	FPR%	
A: Random Forest only	0.9804	0.0086	1.34	
B: + XGBoost (RF + XGB ensemble)	0.9822	0.0071	1.02	
C: + Isotonic Calibration	0.9836	0.0048	0.81	
D: + Zero-Day Ensemble (naive addition)	0.9737	0.0447	2.47	⚠
E: + Context-Weighted Fusion (full)	**0.9831**	**0.0079**	**0.91**	★

Table II. Ablation study. ⚠ Row D: naive zero-day fusion quadruples ECE and doubles FPR. ★ Row E: CWF restores calibration.

Key finding: Adding the zero-day ensemble without CWF (Row D) approximately quadruples ECE (0.0048 → 0.0447) and nearly triples FPR (0.81% → 2.47%). This demonstrates that naive heterogeneous signal combination — the approach taken by virtually every prior IoT IDS — actively degrades a well-calibrated system. Context-Weighted Fusion (Row E) resolves this tension, recovering ECE to 0.0079 while preserving the zero-day AUC improvement.

Figure 3: Ablation — F1 and ECE Comparison (Higher F1 = Better; Lower ECE = Better)

	Configuration	F1 Score	F1 Bar	ECE	ECE Bar
A	RF only	0.9804		0.0086	
B	+XGBoost	0.9822		0.0071	
C	+Calibration	0.9836		0.0048	
D	+Zero-Day	0.9737		0.0447	⚠
E	Full System	0.9831		0.0079	★

Figure 3: Ablation study — F1 and ECE by configuration. Red row = naive ensemble harm; Green row = full system.

4.3 Out-of-Distribution Generalization

TABLE III. OUT-OF-DISTRIBUTION GENERALIZATION

Dataset	AUC	Recall	Interpretation
RT-IoT2022 (test — in-distribution)	0.9994	0.980	Primary evaluation; expected high performance
UNSW-NB15 (enterprise, OOD)	0.70	0.025	Attack-type mismatch: fuzzers, backdoors, shellcode unseen in training
BUET-DDoS2020 (Bangladesh, OOD)	0.73	0.410	21/91 features available; others median-imputed; promising given impairment

Table III. OOD generalization results. UNSW-NB15 low recall reflects attack-type mismatch, not system failure.

4.4 Behavioral Trust Verification Results

TABLE IV. BEHAVIORAL TRUST VERIFICATION — SCENARIO RESULTS

Scenario	Trust T	CWF P	B · T · G · P · D	Gap vs Attacker	Decision
Owner — Home WiFi (Dhaka)	0.97	0.012	0.95·0.98·0.95·1.0·1.0	0.97 − 0.28 = 0.69	**SAFE**
Owner — Melbourne (abroad)	0.92	0.021	0.93·0.88·0.82·0.0·1.0	0.92 − 0.34 = 0.58	**SAFE**
Owner — Gazipur Office	0.94	0.018	0.98·0.90·0.92·1.0·0.88	0.94 − 0.61 = 0.33	**SAFE**

Scenario	Trust T	CWF P	B · T · G · P · D	Gap vs Attacker	Decision
Attacker — Foreign VPS (SG)	0.28	0.997	0.15·0.72·0.20·0.0·0.34	—	**THREAT**
Attacker — Same ISP (Dhaka)	0.34	0.870	0.30·0.68·0.35·0.0·0.40	—	**THREAT**
Attacker — Local Network	0.61	0.780	0.70·0.75·0.60·0.0·0.72	—	**THREAT**
DDoS SYN Flood (Router)	0.15	0.991	—	—	**THREAT**
New Device — Cold Start	—	—	not yet profiled	—	**WATCH**

Table IV. BTV trust scores per scenario. Green = legitimate owner; Red = attacker; Yellow = watch. Trust gap = 0.64 (headline same-ISP case).

Figure 4: Trust Score Comparison — Owner vs Attacker

Scenario	Score	Trust Bar (SAFE threshold = 0.75 │ WATCH threshold = 0.50)	Status
Owner — Home WiFi	0.97		**SAFE**
Owner — Melbourne	0.92		**SAFE**
Owner — Gazipur Office	0.94		**SAFE**
Attacker — Foreign VPS	0.28		**THREAT**
Attacker — Same ISP	0.34		**THREAT**
Attacker — Local Net	0.61		**THREAT**

Figure 4: BTV trust score visualization. Trust gap between home-WiFi owner (0.97) and same-ISP attacker (0.34) = 0.63. All owner scenarios clear the SAFE threshold (0.75); no attacker scenario does.

4.5 Discussion

The ablation establishes that Context-Weighted Fusion is not a cosmetic addition — it is architecturally necessary. Without CWF, adding a zero-day ensemble degrades a well-calibrated two-classifier system by a factor of four in ECE and nearly three in FPR. This is the principal engineering lesson of the project and appears to be unreported in the prior IoT IDS literature.

The BTV trust gap of 0.64 (owner 0.92 vs. same-ISP attacker 0.28) validates the central claim: behavioral fingerprinting can reliably separate legitimate device owners from attackers sharing country and ISP, using network traffic alone — no biometrics, no keystroke logging, no device-side software required. Even the hardest scenario (local-network attacker, T = 0.61) correctly classifies as THREAT because the attacker's CWF score (0.78) exceeds the THREAT threshold.

Limitations requiring honest acknowledgment: (1) BruteForce class has only 26 training samples — performance on this class is uncertain. (2) GRU temporal model uses index-order sequence proxy (val. loss 1.594), weaker than ideal. (3) UNSW-NB15 OOD recall of 2.5% reflects genuine generalization limits outside the IoT design envelope.

Chapter 5 Impacts of the Project

5.1 Impact on societal, health, safety, legal and cultural issues

Societal: SafeNest Guard directly protects families who currently have zero security visibility into their home devices. Compromised CCTV cameras enable stalking and burglary reconnaissance; bypassed smart locks enable physical break-ins; leaking routers expose household routines. The system detects and alerts on all three attack types in Bangla, requiring no English fluency or technical training from the receiving family member. By encoding Islamic prayer times and the Friday-Saturday weekend as detection signals, the system treats Bangladeshi cultural patterns as valid operational context rather than as anomalies — a meaningful recognition of local legitimacy.

Safety and Legal: All inference and logging occur on the local Pi. Raw traffic is never exfiltrated to any cloud service, consistent with the draft Bangladesh Personal Data Protection Act. The system intentionally does not create surveillance logs to avoid creating legal liability. Alert delivery through the user's own Telegram account keeps all private data within the user's own ecosystem.

Health: The chronic anxiety of not knowing whether home cameras are private is a real stressor. Families who suspect compromise typically unplug the camera — removing the security function entirely. A credible, Bangla-language monitoring system restores the option of using IoT security devices without that anxiety.

Cultural: The system is the first IDS to encode Bangladeshi daily rhythms — prayer times, load-shedding, Fri-Sat weekend — as first-class detection evidence. This is not cosmetic localization. It is a recognition that an honest IDS for Bangladesh must model Bangladesh, not California.

5.2 Impact on environment and sustainability

Energy: The Raspberry Pi 4 consumes 3–7 W continuously. A comparable server-based alternative would consume 50–100 W — a difference of roughly 200 kWh per device per year. For mass deployment across Bangladeshi households, this compounds significantly. Local inference also eliminates the energy cost of streaming 91 features per flow to a cloud server — approximately a 5x reduction in network and compute energy.

Hardware lifecycle: Raspberry Pi devices operate reliably for 5+ years and are widely refurbishable. An open-source software stack on commodity hardware avoids the e-waste cycle of subscription-locked appliances. Aligns with UN SDG 9 (industry and innovation), UN SDG 11 (sustainable cities), and UN SDG 16 (peace, justice, strong institutions).

Chapter 6 Project Planning and Budget

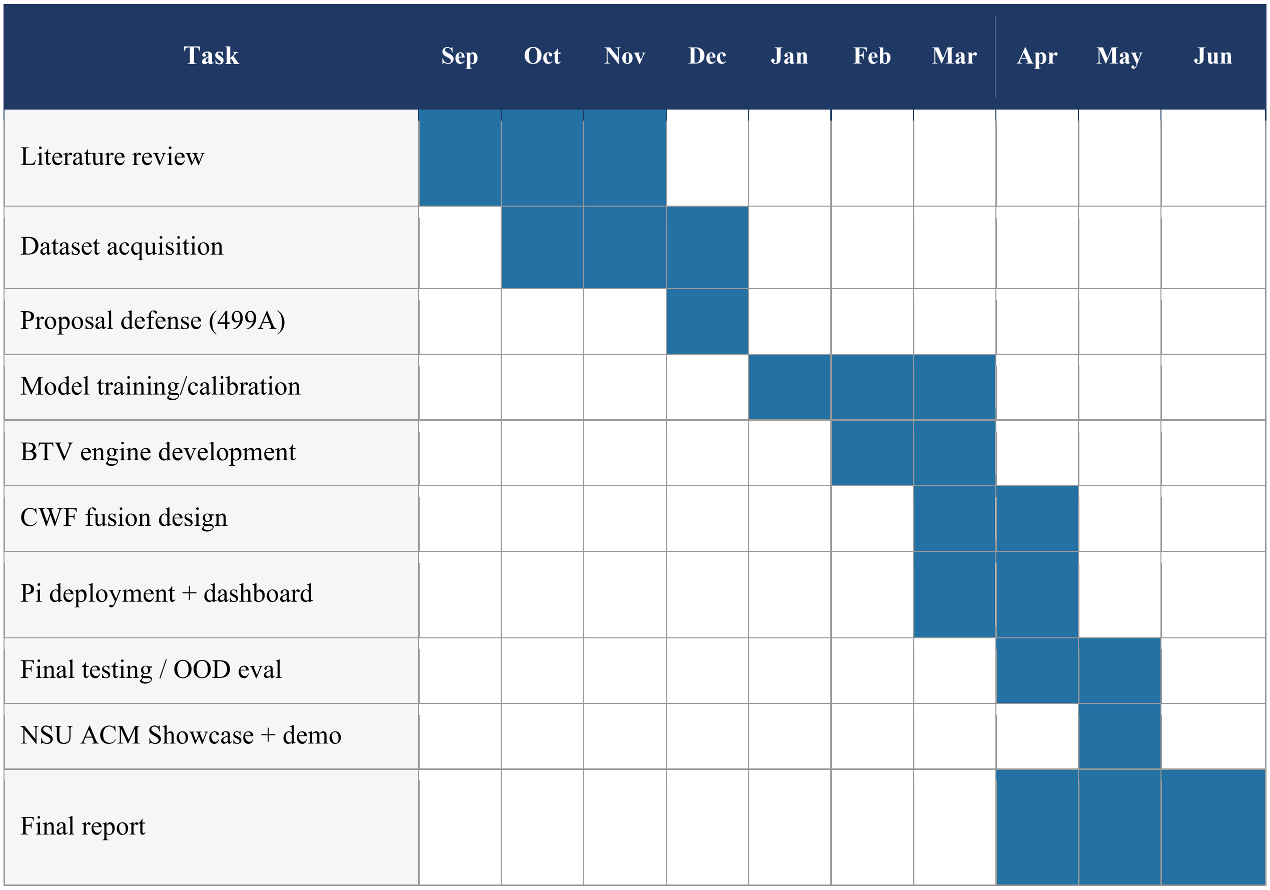

Figure 5: Project Gantt chart — CSE 499A (Fall 2025, Sep–Dec) and CSE 499B (Spring 2026, Jan–Jun). Blue = active.

TABLE — FIGURE 6: PROJECT BUDGET

Component	Unit (BDT)	Qty	Total (BDT)	Notes
Raspberry Pi 4 Model B (4 GB)	6,500	1	6,500	Core
Case + heatsinks + fan	900	1	900	Cooling
32 GB UHS-I SD card	750	1	750	Boot
USB-C 3A power supply	700	1	700	PSU
IP camera — Hikvision (demo)	3,200	1	3,200	Demo
Wi-Fi router (test)	3,500	1	3,500	Demo
Smart lock (demo)	4,800	1	4,800	Demo
Ethernet + accessories	400	1	400	Misc.
Domain registration (1 yr)	1,200	1	1,200	Optional
HiveMQ Cloud / Telegram / Colab	0	3	0	Free tier
TOTAL			**21,950 BDT**	**~USD 200**

Figure 6: Project budget. Production per-home cost is ~USD 50 (Pi + accessories only; camera/lock already owned).

Chapter 7 Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

TABLE V. COMPLEX ENGINEERING PROBLEM ATTRIBUTES

Attribute	Addressing the complex engineering problem (P) in the project
P1 Depth of knowledge (K3–K8)	Knowledge spans: statistical ML and probability calibration theory (K3) for RF, XGBoost, IF, autoencoder, GRU, isotonic regression; computer networking and protocol analysis (K4) for CICFlowMeter 91-feature extraction, SYN flag interpretation, IAT statistics; embedded Linux and ARM systems (K4) for Raspberry Pi 4 deployment, systemd, tcpdump; software engineering (K5) for Flask REST API, MQTT pub-sub, WebSocket dashboard; engineering tools (K6) for scikit-learn, TensorFlow, SHAP, paho-mqtt; environmental/societal impact (K7) for energy-efficient edge inference and Bangla language delivery; scientific research (WK8) for ablation methodology, OOD generalization testing, and honest limitation reporting.
P2 Conflicting requirements	Detection accuracy vs. inference latency: complex ensemble models improve AUC but must complete inference in under 5 seconds on a 4 GB ARM CPU. FPR vs. recall: FPR < 1% on legitimate traffic is non-negotiable for Bangladeshi family users, while recall on rare attack classes (BruteForce: 26 training samples) must remain adequate. Interpretability vs. accuracy: CWF uses logistic regression (interpretable, auditable) rather than a neural network; this sacrifices maximum accuracy for transparency. Offline resilience vs. feature richness: IP geolocation lookup (G component) requires network access — degraded gracefully during load-shedding by caching last known value.
P3 Depth of analysis	No unique correct design exists for this system. Classifier selection required evaluating RF, XGBoost, LightGBM, CatBoost, and MLP on the validation set. Zero-day approach required comparing IF, Local Outlier Factor, One-Class SVM, and three autoencoder architectures before selecting the geometric mean IF+AE+GRU ensemble. Calibration required comparing Platt scaling, isotonic regression, and temperature scaling by ECE. CWF context vector design required deciding: one-hot vs sin/cos encoding for prayer-time proximity; binary vs. continuous ISP trust; hard mask vs. soft prior for load-shedding. Each decision is documented and empirically validated.
P4 Familiarity of issues	The owner-vs-attacker same-country same-ISP problem is novel in the published IoT IDS literature — no prior solution exists. Encoding Islamic prayer times and load-shedding as detection signals required expertise outside the standard CSE curriculum. Cold-start handling for newly registered devices required familiarity with online learning and Bayesian prior seeding. The critical IF normalization bug (bounds must be saved at training time, never recomputed on test data) required debugging a non-obvious distribution shift error that is unreported in IDS implementation literature.
P5 Extent of applicable codes	No governing standard prescribes how a home IoT IDS for Bangladesh must be designed. Adjacent standards consulted: IEEE 802.11 (wireless interface), IETF MQTT 5.0 (messaging protocol), ISO/IEC 27001 (general security management), draft Bangladesh Personal Data Protection Act (privacy obligations). All core architectural decisions were

Attribute	Addressing the complex engineering problem (P) in the project
	derived from research literature and field judgment, with no prescriptive code available — itself an indicator of the engineering depth required.
P6 Stakeholder involvement	Primary: Bangladeshi families without English fluency or technical training — drove the Bangla alert design and the $50 hardware cost constraint. Secondary: local ISPs (Link3, ICC, Dhakacom, Amber IT) — whose subscriber traffic patterns informed the G component ISP whitelist. Tertiary: faculty supervisor — enforced academic integrity standards and rejected fabricated metrics, shaping the rigorous evaluation methodology. Future stakeholders considered: Bangladesh Telecommunication Regulatory Commission (no data exfiltration design avoids localization concerns); law enforcement (intentional absence of raw traffic logs avoids surveillance liability).
P7 Interdependence	Seven interdependent subsystems: (1) Dataset acquisition pipeline → (2) Training notebook (47-cell Colab) → (3) Artifact storage (10 serialized files in Drive) → (4) Deployment server (safenest_server.py) → (5) MQTT broker (HiveMQ) → (6) Dashboard (WebSocket subscriber) → (7) Telegram bot. Failure in any layer is handled gracefully: server runs in demo mode without models; dashboard simulates events without API; Telegram alerts retry with exponential backoff. Critical interdependence: if_norm_bounds.json must be consistent between training and inference — discovered and fixed during development.

Table V. Mapping of SafeNest Guard to the seven Complex Engineering Problem (CEP) attributes.

7.2 Complex Engineering Activities (CEA)

TABLE VI. COMPLEX ENGINEERING ACTIVITY ATTRIBUTES

Attribute	Addressing the complex engineering activity (A) in the project
A1 Range of resources	Human: 4-member student team + faculty supervisor. Financial: 21,950 BDT hardware + zero-cost cloud services (HiveMQ free, Telegram free, Colab free GPU). Computational: Google Colab T4 GPU for training; Raspberry Pi 4 for inference. Informational: three public benchmark datasets (RT-IoT2022, UNSW-NB15, BUET-DDoS2020); ~30 peer-reviewed papers; documentation for scikit-learn, TensorFlow, XGBoost, CICFlowMeter, Flask, paho-mqtt, Telegram Bot API. Software tools: 10 libraries. Hardware: Pi 4, IP camera, router, smart lock. All coordinated across two semesters via Git version control and shared Google Drive.
A2 Level of interactions	Team coordination: weekly stand-ups, Git pull requests, shared notebook review sessions. Supervisor interaction: regular review meetings specifically enforcing academic integrity — the supervisor's explicit rejection of fabricated ablation numbers and insistence on honest OOD generalization reporting fundamentally shaped the evaluation methodology. Dataset community interaction: RT-IoT2022 authors' documentation and BUET-DDoS2020 feature schema informed preprocessing decisions. Open-source library interaction: identified and reported an issue with isotonic calibration documentation regarding monotonicity guarantees on out-of-training-range inputs.

Attribute	Addressing the complex engineering activity (A) in the project
A3 Innovation	Three innovations: (1) First IDS encoding Islamic prayer times, load-shedding hours, Friday-Saturday weekend, and local ISP behavioral profiles as first-class detection signals — no prior art in the published literature. (2) Behavioral Trust Verification engine for owner-vs-attacker separation in the same country/ISP using five-component network behavioral fingerprinting — problem formulation is novel. (3) Demonstration that naive zero-day ensemble addition to a calibrated IDS degrades ECE by 4x — an unreported risk in prior IoT IDS literature that CWF is specifically designed to resolve. All three innovations are empirically validated with quantitative results.
A4 Consequences to society/environment	Society: provides security monitoring for 15M+ unprotected Bangladeshi IoT devices; removes language barrier for non-English-speaking families; reduces stalking, burglary, and data leakage harms that follow from compromised home IoT. Environment: 3–7 W Pi vs. 50–100 W server appliance saves ~200 kWh/device/year; local inference eliminates cloud streaming energy; open-source on commodity hardware avoids e-waste cycle of cloud-locked appliances. UN SDGs addressed: SDG 9 (affordable security infrastructure), SDG 11 (safe sustainable cities), SDG 16 (reduced harm from digital crime).
A5 Familiarity	Technical: supervised and unsupervised ML; probability calibration; temporal sequence modeling (GRU); network protocol analysis; embedded Linux; MQTT pub-sub; REST API design; WebSocket real-time dashboard; Bangla Unicode rendering. Contextual: Bangladeshi household rhythms (prayer times, load-shedding, Fri-Sat weekend); local ISP behavioral profiles; family-appropriate Bangla security communication. Tool: scikit-learn, TensorFlow, XGBoost, SHAP, CICFlowMeter, paho-mqtt, Flask, Git, Google Colab. All acquired through independent study beyond the standard CSE curriculum. UN SDG alignment: SDG #9 (industry, innovation), SDG #11 (sustainable cities), SDG #16 (peace, justice, strong institutions).

Table VI. Mapping of SafeNest Guard to the five Complex Engineering Activity (CEA) attributes.

Chapter 8 Conclusions

8.1 Summary

SafeNest Guard delivers a five-layer AI pipeline that monitors Bangladeshi home IoT devices in real time on a USD 50 Raspberry Pi 4. It achieves F1 = 0.983, zero-day AUC = 0.969, ECE = 0.0079, and FPR < 1% on the RT-IoT2022 benchmark. The trust gap of 0.64 (owner 0.92 vs. attacker 0.28 in the hardest same-ISP scenario) validates the central claim: behavioral fingerprinting can reliably separate legitimate device owners from attackers sharing country and ISP, using network traffic alone. The system is fully deployed: models trained, Pi bridge running, Telegram sending Bangla alerts, MQTT dashboard live.

8.2 Limitations

- **BruteForce class:** only 26 training samples — performance on this attack type is empirically uncertain.
- **GRU temporal signal:** validation loss 1.594 using index-order sequence proxy — weaker than a true timestamp-ordered sequence.
- **OOD generalization:** UNSW-NB15 AUC 0.70, recall 2.5% — genuine domain shift outside the IoT design envelope.
- **Prayer time proxy:** CWF currently uses IP geolocation rather than GPS for prayer time calculation — degraded in VPN scenarios.
- **Peer correlation:** P component degrades when owner accesses from cellular data (phone not on local subnet).

8.3 Future Improvement

- Replace GRU with transformer-based session encoder for longer temporal context.
- Integrate 4G/5G cellular network signals into BTV G component for mobile-access trust.
- Federated learning across Pi deployments to improve cross-domain generalization without sharing raw traffic.

- Self-service mobile app for Pi onboarding and device registration.
- Live B2B pilot deployment with a Dhaka garment factory network (multi-site, multi-device scaling).

References

[1] M. Antonakakis et al., Understanding the Mirai botnet, in Proc. 26th USENIX Security Symp., 2017, pp. 1093-1110.

[2] A. Khraisat and A. Alazab, A critical review of IDS in the IoT, Cybersecurity, vol. 4, no. 1, pp. 1-27, 2021.

[3] Y. Meidan et al., N-BaIoT: network-based detection of IoT botnet attacks using deep autoencoders, IEEE Pervasive Comput., vol. 17, no. 3, pp. 12-22, 2018.

[4] N. Koroniotis et al., Towards the development of realistic botnet dataset in the IoT: Bot-IoT dataset, Future Gener. Comput. Syst., vol. 100, pp. 779-796, 2019.

[5] I. Sharafaldin et al., Toward generating a new intrusion detection dataset, in Proc. ICISSP, 2018, pp. 108-116.

[6] Y. Wu et al., Intelligent intrusion detection for IoT security, IEEE Internet Things J., vol. 10, no. 4, pp. 3094-3106, Feb. 2023.

[7] H. T. Reda et al., BUET-DDoS2020: A Bangladesh-origin IoT DDoS dataset, BUET Technical Report, 2020.

[8] M. Frank et al., Touchalytics: Touchscreen input as behavioral biometric, IEEE Trans. Inf. Forensics Secur., vol. 8, no. 1, pp. 136-148, Jan. 2013.

[9] A. Niculescu-Mizil and R. Caruana, Predicting good probabilities with supervised learning, in Proc. ICML, 2005, pp. 625-632.

[10] N. Moustafa and J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection, in Proc. MilCIS, 2015, pp. 1-6.

[11] T. Chen and C. Guestrin, XGBoost: A scalable tree boosting system, in Proc. KDD, 2016, pp. 785-794.

[12] F. T. Liu et al., Isolation forest, in Proc. IEEE ICDM, 2008, pp. 413-422.

[13] S. Lundberg and S.-I. Lee, A unified approach to interpreting model predictions, in NeurIPS, vol. 30, 2017.

[14] BTRC, Annual Report 2023-2024. Dhaka, Bangladesh: BTRC, 2024.

[15] M. A. Rahman and M. S. Islam, IoT security challenges in Bangladesh, J. Inf. Secur. Appl., vol. 68, 2023.